

Movie Genre Classification

Professionalized machine learning repository report

Problem

Predict movie genres from plot descriptions using text classification.

Workflow

Raw dataset -> preprocessing -> feature engineering -> model comparison -> evaluation outputs -> saved model -> Streamlit app.

Models Compared

Model	Purpose
Naive Bayes	Baseline comparison candidate
Logistic Regression	Baseline comparison candidate
Linear SVM	Baseline comparison candidate

Dataset Columns

Column
id
title
genre
description
clean_description

Repository Improvements

- Organized folders for data, notebooks, source code, models, outputs, app, docs, images, and reports.
- Moved reusable logic into modular Python files.
- Added config.yaml for reproducible parameters.
- Added logging-oriented training scripts and output artifacts.
- Added documentation for business context, architecture, data dictionary, deployment, and usage.

Key Notes

- The model predicts top genre candidates from a synopsis.
- Macro F1 is used because genre classes can be imbalanced.

Next Steps

- Run training after installing dependencies.
- Capture Streamlit screenshots for the images folder.
- Add SHAP or feature contribution explanations if presentation time allows.